

Kripesh Nihure

AI/ML Engineer with experience building domain-adapted LLM systems, NLP pipelines, and production-ready machine learning applications. Skilled in transformer fine-tuning, corpus curation, prompt engineering, and end-to-end ML workflows from data collection to deployment.

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WORK EXPERIENCE

Wiseyak Solutions

AI Engineer

Mar 2026 - Present

- Architecting and developing a multi-tenant Retrieval-Augmented Generation (RAG) platform that enables organizations to deploy domain-specific AI assistants using their own knowledge bases, prompts, and business workflows.
- Designing hybrid retrieval systems combining semantic search, keyword search, reranking, and knowledge graph techniques to improve answer relevance and reduce hallucinations in production AI applications.
- Building end-to-end AI pipelines covering document ingestion, preprocessing, chunking, embedding generation, vector indexing, retrieval, reranking, and response generation for enterprise-scale knowledge management.
- Developing configurable SaaS infrastructure that allows organizations to customize AI behavior, retrieval strategies, prompt templates, and domain-specific workflows without modifying core platform code.
- Contributing to a real-time Voice-to-Voice (V2V) conversational AI platform that enables organizations to automate high-volume phone interactions using speech recognition, LLM orchestration, and natural voice synthesis.
- Engineering low-latency conversational pipelines integrating VAD, streaming STT/TTS, contextual memory, and intelligent agent workflows to deliver human-like voice interactions over telephony channels.

Kavaya Research

Data scientist and AI/ML Engineer

May 2025 - Feb 2026

- Built domain-adapted NLP/LLM systems using proprietary datasets collected from users monitored in-house across 1 year of social media activity.
- Designed and optimized corpus curation, preprocessing, tokenization, and labeling pipelines for large-scale transformer training workflows.
- Performed domain-adaptive pretraining and fine-tuning of transformer models, improving downstream performance by 3-5%.
- Developed LLM-assisted annotation pipelines to generate high-quality supervised training datasets at scale.
- Built and evaluated ML models across structured and unstructured datasets using PyTorch, HuggingFace Transformers, and Scikit-learn.
- Conducted feature engineering, experimentation, and model evaluation for predictive and NLP tasks achieving up to 94% accuracy.

Adex International

Cloud Apprentice

Mar 2025 - April 2025

- Worked with AWS services and cloud fundamentals for deploying and managing scalable systems
- Gained hands-on exposure to cloud architecture, storage, and compute services
- Preparing for AWS Solutions Architect Associate certification

EDUCATION

Pulchowk Campus, IOE

Bachelor in Computer Engineering - 80.25%

Feb 2021 - Feb 2025

- **Relevant Coursework:** Artificial Intelligence, Data Structures & Algorithms, Database Management Systems, Discrete Structures, Probability & Statistics, Object Oriented Programming, Software Engineering
- **Samsung Innovation Campus** – Applied Data Science, awarded as an Excellent Student
- DevOps Elective: Docker, Kubernetes, CI/CD, AWS

CERTIFICATE

Associate Data Scientist

- Completed hands-on learning track of 33 data science courses covering Python, data manipulation, supervised & unsupervised learning, statistics, regression, hypothesis testing, and exploratory data analysis.

Solutions Architect Associate (SAA-C03)

- Learnt leveraging AWS best practices to design resilient, enterprise-grade architectures, optimizing for both technical performance and operational spend.

PROJECTS

NLP based DBMS: Natural language interface to DBMS

- Built an end-to-end NLP system to convert natural language queries into SQL queries
- Fine-tuned a T5 transformer model, improving BLEU score from 0.15 to 0.80
- Designed data preprocessing and training pipelines for structured query generation
- Developed a full-stack application (React + Python) and deployed using CI/CD on Azure

DermaDetector: AI-based skin disease classifier

- Developed a deep learning model (EfficientNet) to classify 15+ skin conditions with 84% accuracy
- Integrated TinyLlama for personalized treatment suggestions and prompt-based recommendations
- Won **Promising Idea** award for innovative real-time skin condition analysis at the Viveka hackathon 2024

JhattaSamachar: Real-time news summarizer

- Built an NLP pipeline to scrape, preprocess, and summarize real-time news
- Fine-tuned T5 model improving ROUGE score from 0.39 to 0.54
- Designed automated data pipeline for continuous content processing

Plagiarism checker: Document similarity comparison tool

- Developed document similarity system using NLP and cosine similarity
- Implemented n-gram based comparison for improved detection accuracy
- Built full-stack interface for document upload and evaluation

SKILLS

- **Programming & Tools:** React, Flask, Django, HTML/CSS, JavaScript, Axios, Firebase, Python, SQL, Docker, Kubernetes, AWS, Git, GitHub, Canva, Figma
- **Data Science & Machine Learning:** Supervised & Unsupervised Learning, Regression, Hypothesis Testing, Experimental Design, NLP, TensorFlow, Keras
- **Data Analysis & Processing:** Pandas, Numpy, Seaborn, Matplotlib, Data Cleaning, Statistical Analysis